Tony Chen

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EDUCATION

The University of Texas at Austin Bachelor of Science, Electrical and Computer Engineering August 2024 – May 2027
Current GPA: 3.9/4.0

Relevant Coursework: Embedded Systems, Circuit Theory, Software Design and Implementation I/II, Digital Logic Design, Linear Systems and Signals, Computer Architecture, Circuits Laboratory, Algorithms

EXPERIENCE

Control Systems Engineer Intern – *KBR, Inc.; Houston, Texas* May 2025 – August 2025

- Designed **control, process, and instrumentation diagrams** for pre-FEED phase of LYB pyrolysis oil project. Performed LOPA re-assessments for Proman ammonia plant, contributing to updated risk-reduction recommendations.
- Investigated remote instrument enclosure manufacturing technologies on-site for clients and developed industrial cybersecurity specifications, **improving client confidence** in enclosure feasibility and secure system architecture.
- Presented the results of my internship project to the Controls Engineering Department and delivered technical presentations during monthly departmental meetings.

High Voltage Lead – *Longhorn Racing Solar* Fall 2024 – Present

- Led high voltage subsystem to develop electronics and firmware for **intelligent, safety-critica**l power systems of a solar-powered vehicle with the goal of optimizing power efficiency, high voltage safety, and rapid design to integration.
- Developed new projects including the Vehicle Control Unit, Active Precharge Board, and Leader System-On-Module. Also oversaw completion of Charger Interface and LV Carrier Base by reviewing designs, guiding debugging, and system integration.
- Designed **multi-layer PCBs** with STM32 and ESP32 microcontrollers in KiCad, developed C/C++ **RTOS firmware** for motor control/power distribution, and assembled HV wire harness.

Research Assistant – *The University of North Texas; Denton, Texas* January 2023 – December 2023

- Analyzed **microprocessor architecture** side-channel vulnerabilities under Dr. Fernando Mosquera and Dr. Krishna Kavi to identify and prevent potential novel cybersecurity attack vectors focusing on CPU memory cache design.
- Developed Gem5 simulations of Spectre and other speculative execution attacks on CPU cache architecture for x86 and ARM to evaluate cache behavior under adversarial conditions, catching over **96%** of potential cache hit attacks.

PROJECTS, LEADERSHIP EXPERIENCE, AND ACTIVITIES

Analog Hall-Effect Keyboard Spring 2026

- Designed and built a custom analog keyboard using an STM32 microcontroller. Implemented **analog signal filtering** for Hall-effect sensors, modeling magnetic field strength vs. voltage response to achieve precise key actuation detection within 0.1mm, and integrated an SPI-driven display for **real-time feedback and system diagnostics**.
- Developed firmware using FreeRTOS, implementing **task scheduling** for sensor polling, display updates, and USB communication. Configured and optimized ADC, SPI, and USB peripherals, including DMA usage and interrupt handling.

Inverted Pendulum Reaction Wheel System Summer 2025

- Designed and programmed a reaction-wheel inverted pendulum using an Arduino microcontroller, reading angular velocity data from a gyro IMU over I2C, and implementing a turned PID control algorithm.
- Modeled **system dynamics**, tuned PID gains experimentally, and optimized loop timing to achieve stable real-time balance resistance to disturbance conditions.

Technical Skills

Skills: PCB Design, Embedded Software Development, FEED, Controls Engineering
Software Tools: C/C++, Java, Python, Verilog, Assembly, FreeRTOS, Linux, Vim, Git
Hardware Experience: KiCad, LTSpice, CAD (Onshape, AutoCAD), Soldering, and Wire Harnessing
Eligible to work in the US without restriction